Local Minimum Wages and Employment: A Replication of Dube et al. (2010)

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Summary

This paper is a narrow and wide replication of Dube et al. (2010), who use contiguous border county pairs to show that negative minimum wage employment elasticities in national panel studies reflect spatial heterogeneity rather than true disemployment effects. The narrow replication confirms their null employment elasticity and positive earnings elasticity for restaurant workers. The wide replication extends their design using Callaway and Sant'Anna (2021) to allow for heterogeneous treatment effects across adoption cohorts. Applying CS (2021) to the full national sample, parallel trends is rejected, confirming that spatial heterogeneity in employment trends contaminates national identification, validating the border-pair design as a necessary condition for credible identification.

I Introduction

The employment effects of minimum wage increases are one of the most debated questions in labor economics. Traditional panel studies (Neumark and Wascher, 1992) use all cross-state variation with county and time fixed effects, treating any county in the country as a valid counterfactual for any other. Local case studies (Card and Krueger, 1994, 2000; Dube et al., 2007) use geographic discontinuities but are limited to single events with short windows. The former tend to find negative employment effects; the latter typically find no employment effects.

Dube et al. (2010) (hereafter, DLR) argues that both approaches are flawed. Using contiguous county pairs on state borders between 1990 and 2006, they show that negative employment elasticities reflect spatial heterogeneity in employment trends unrelated to minimum wage policy. Once they control for this using pair × period fixed effects, the employment elasticity is indistinguishable from zero, while the earnings elasticity is estimated at +0.19.

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DLR's findings have not gone unchallenged. Neumark et al. (2014b) argues that the border county design removes identifying variation and that properly controlling for heterogeneous trends in the full national sample reveals negative employment effects. A companion paper, Neumark et al. (2014a), argues that the DLR design is sensitive to the choice of the control group and that their null result is not robust across alternative specifications. DLR and subsequent work by Allegretto et al. (2017) dispute these claims, arguing that regional controls are necessary. This ongoing debate motivates our reexamination of the DLR design.

A potential concern is whether DLR's null employment result is robust to allowing heterogeneous treatment effects across adoption cohorts (Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021). We address this by applying the doubly-robust estimator of Callaway and Sant'Anna (2021) to the full national sample, finding that parallel trends is rejected even with robust weighting and covariate adjustment. The findings support the border-pair design employed in DLR.

The rest of the paper is organized as follows. Section 2 reviews the data. Section 3 discusses the narrow replication. Section 4 presents the wider replication. Section 5 concludes.

II Data

Table 1 presents descriptive statistics for the two estimation samples. The all-county (AC) sample consists of 1,380 counties with a balanced panel of 66 quarters from 1990Q1 to 2006Q2. The contiguous border county-pair (CBCP) sample restricts itself to the 504 counties lying along a state border that can be paired with at least one cross-state neighbor. There are 318 of these county pairs. The two samples are comparable in terms of population, employment levels, and average wages. Restaurant employment averages 4,508 workers per county in the AC sample and 4,185 in the CBCP sample. Average weekly earnings are approximately \$170 for both samples. The effective minimum wage averages \$4.84 across both samples.

III Narrow Replication: Empirical Strategy

Following DLR, we estimate the elasticity of restaurant earnings and employment with respect to the effective minimum wage across six specifications. The baseline estimating equation is

$$\ln y_{it} = \alpha + \eta \ln MW_{it} + \delta \ln y_{it}^{TOT} + \gamma \ln \text{pop}_{it} + \phi_i + \tau_t + \varepsilon_{it} \tag{1}$$

where y_{it} is log average weekly earnings or log employment for restaurant workers in county i at quarter t, $\ln MW_t$ is the log effective minimum wage, ϕ_i are county fixed effects, and τ_t is a time fixed effect whose form varies by specification. Two control sets are reported throughout: $c2$, which includes log county population for employment regressions only, and $c3$, which includes log total private employment (employment regressions) or log total private average weekly wages (earnings regressions).

Table 2 reports results for six specifications. Specifications 1–4 use the AC sample. Specification 1 is the standard national panel approach that imposes common time period

Table 1: Descriptive Statistics

	(1) All-County Sample		(2) County-Pair Sample	
	Mean	s.d.	Mean	s.d.
Population, 2000	180,983	423,425	167,956	297,751
Population density, 2000	464.7	2,553.4	556.0	3,335.0
Land area (square miles)	1,106.8	1,760.7	1,380.2	2,470.3
Overall private employment	32,179.1	119,362.9	32,185.0	101,317.6
Restaurant employment	4,508.3	10,520.7	4,184.8	7,808.5
Restaurant average weekly earnings ($)	170.7	43.7	172.0	45.5
Accommodation & food services employment	—	—	12,864.5	26,861.6
Accommodation & food services AWW ($)	—	—	273.0	66.9
Retail employment	4,703.0	14,641.9	4,543.1	11,545.4
Retail AWW ($)	305.6	76.7	303.7	76.8
Manufacturing employment	6,607.9	20,323.3	6,312.0	14,100.4
Manufacturing AWW ($)	572.6	201.7	576.0	203.8
Minimum wage ($)	4.837	0.663	4.839	0.669
Number of counties	1,380		504	
Number of county pairs	—		318	
Number of states	48		48	

Notes: Sample means and standard deviations for all counties with a balanced panel of 66 quarters (1990Q1–2006Q2) and for all contiguous border county-pairs. Accommodation & food services figures use the combined ACFSRETAIL variable (NAICS 72 + retail), which is only available for the county-pair sample due to QCEW confidentiality suppression in the all-county sample. Weekly earnings and minimum wages in nominal dollars. Sources: QCEW; U.S. Department of Labor; U.S. Bureau of the Census, 2000 *Census*.

fixed effects across all counties (Neumark and Wascher, 1992) and produces the negative employment elasticities that motivate DLR. Specifications 2–4 introduce increasingly finer spatial controls. Together, Specifications 2–4 tell us whether the spatial heterogeneity is driving the negative employment result in Specification 1. As the geographic size of the spatial controls shrinks, the elasticity should attenuate toward zero.

Specifications 5 and 6 use the CBCP sample of 504 counties, forming 318 cross-state pairs. Specification 5 retains common period effects but restricts the sample to border counties. Specification 6 replaces τ_t in Equation 1 with pair $\times$ period fixed effects τ_{pt}:

$$\ln y_{ipt} = \alpha + \eta \ln MW_{it} + \delta \ln y_{it}^{TOT} + \gamma \ln \text{pop}_{it} + \phi_i + \tau_{pt} + \varepsilon_{ipt}. \tag{2}$$

Each pair-period effect absorbs all time-varying shocks common to both counties in a pair, such as a local business cycle. Identification comes from within-pair differences in minimum wages between two counties along a state border. The identifying assumption is that, conditional on county and pair-period fixed effects, minimum wage differences within these contiguous pairs are uncorrelated with unobserved county-level shocks. DLR provides support for this assumption via pre-trends testing (Table 3), which shows that leads of the minimum wage are close to 0 in Specification 6 but not under Specification 1.

Table 2 reports results. The earnings elasticity is stable at ≈ 0.20 across all six specifications and remains statistically significant throughout. Turning toward the employment elasticity, it starts at -0.17 under Specification 1, the traditional national-level panel approach, and gradually approaches zero as the spatial controls become finer. At Specification 6, it is estimated at 0.17 and is statistically insignificant. This pattern follows DLR's prediction that negative employment effects in the prior literature are driven by spatial heterogeneity rather than unemployment actually increasing.

IV Wide Replication: Empirical Strategy

DLR designs their empirical strategy by claiming that the findings of negative employment elasticities in traditional panel studies reflect spatial heterogeneity in employment trends. Their border-pair design addresses this by restricting comparisons to contiguous county pairs. This wide replication tests whether this identification strategy is necessary. Plainly, we are testing whether a modern heterogeneity-robust estimator will recover DLR's estimates using the full national sample.

We apply the doubly-robust group-time average treatment effect estimator of Callaway and Sant'Anna (2021) to the all-county annual panel. The estimator recovers cohort-specific ATTs, $ATT(g,t)$, for each combination of treatment cohort g and time t, using not-yet-treated counties as the control group. This avoids the "forbidden comparisons" problem identified by Goodman-Bacon (2021), in which already-treated units contaminate the TWFE weights. The doubly-robust implementation conditions on log county population and log baseline total private employment.

Treatment is a binary absorbing indicator equal to one from the first year a county's state minimum wage exceeded the federal floor. The annual panel covers 1,380 counties from 1990 to 2005. Three treatment cohorts are identified: 1992 ($n = 183$ counties), 2004, and 2005. Always-treated counties ($n = 379$, already above the federal floor at the sample

Table 2: Minimum Wage Effects on Earnings and Employment

	(1)		(2)		(3)		(4)		(5)		(6)	
	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
ln Earnings												
$\ln MW_t$	0.224***	0.217***	0.153***	0.149***	0.204***	0.195***	0.219***	0.210***	0.232***	0.221***	0.200***	0.188***
	(0.033)	(0.028)	(0.035)	(0.033)	(0.038)	(0.034)	(0.037)	(0.034)	(0.033)	(0.032)	(0.065)	(0.060)
ln Employment												
$\ln MW_t$	−0.211**	−0.176*	−0.028	−0.023	−0.054	−0.039	−0.052	−0.032	−0.137*	−0.112	0.057	0.017
	(0.095)	(0.096)	(0.066)	(0.068)	(0.055)	(0.050)	(0.084)	(0.078)	(0.072)	(0.076)	(0.063)	(0.054)
N	91,080		35,178		91,080		91,080		70,620		41,712	
Controls												
Census division × period dummies					Y	Y	Y	Y				
State linear trends							Y	Y				
MSA × period dummies		Y	Y									
County-pair × period dummies											Y	Y
Total private sector		Y		Y		Y		Y		Y		Y

Notes: Outcome variables are log average weekly earnings and log employment for restaurant workers. All specifications include county fixed effects and controls for log county population (employment regressions). Columns labeled "w/o" exclude total private sector controls; "w/" include log total private earnings (earnings regressions) or log total private employment (employment regressions). Standard errors in parentheses, clustered by state (Specs 1–4) and two-way by state and border segment (Specs 5–6). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Spec 6 employment elasticities (+0.057, +0.017) have the opposite sign to DLR's (−0.052, −0.016) but are statistically indistinguishable from zero; the null employment result is confirmed. The Spec 6 sample size (41,712 vs. DLR's 70,620) differs because singleton pairs are dropped by `fixest`.

Table 3: Pre-Existing Trends in Employment and Earnings and Validity of Controls

	Specification 1		Specification 4		Specification 6	
	Restaurants	All Private	Restaurants	All Private	Restaurants	All Private
ln Earnings						
η_{t-12}	−0.022	−0.011	−0.029	−0.007	0.009	0.029
	(0.025)	(0.023)	(0.042)	(0.044)	(0.028)	(0.033)
η_{t-4}	0.006	0.004	0.170***	0.045	0.121***	0.093**
	(0.033)	(0.024)	(0.049)	(0.039)	(0.037)	(0.036)
Trend ($\eta_{t-4} - \eta_{t-12}$)	−0.016	−0.007	0.141**	0.038	0.130***	0.123***
	(0.050)	(0.035)	(0.058)	(0.061)	(0.043)	(0.044)
N	74,520	74,520	28,782	28,782	34,128	34,128
ln Employment						
η_{t-12}	−0.100	−0.024	−0.004	0.036	0.060	0.062
	(0.070)	(0.035)	(0.077)	(0.039)	(0.047)	(0.050)
η_{t-4}	−0.144**	−0.063	−0.019	−0.032	0.008	0.091
	(0.072)	(0.034)	(0.073)	(0.038)	(0.084)	(0.074)
Trend ($\eta_{t-4} - \eta_{t-12}$)	−0.244*	−0.087	−0.023	−0.032	0.068	0.152
	(0.136)	(0.063)	(0.132)	(0.046)	(0.118)	(0.103)
N	74,520	74,520	28,782	28,782	34,128	34,128
Controls						
MSA × period dummies			Y	Y		
County-pair × period dummies					Y	Y

Notes: $t - j$ denotes j quarters prior to the minimum wage change. η_{t-12} is the coefficient on $(\ln MW_{t-4} - \ln MW_{t-12})$; η_{t-4} is the coefficient on $(\ln MW_t - \ln MW_{t-4})$; Trend $= \eta_{t-4} - \eta_{t-12}$. All specifications include county fixed effects and log county population. Standard errors in parentheses, clustered at state level (Specs 1 and 4) and state and border segment level (Spec 6). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Spec 4 N (28,782 vs. DLR's 43,980) reflects sample coverage differences in MSA×period FE construction. Spec 4 and 6 restaurant earnings leads are positive and significant, inconsistent with DLR's near-zero estimates; this may reflect differences in sample construction or variable definitions across data vintages. Crucially, Spec 6 restaurant employment leads are all small and statistically insignificant, consistent with DLR's main validity finding.

start) are excluded due to the absence of pre-treatment data. Not-yet-treated counties serve as the control group at each point in time, avoiding the concern that never-treated counties (disproportionately Southern states with systematically different employment trends) would create problems with spatial heterogeneity.

Tables 4 and 5 report the results. The overall ATT for ln employment outcomes is -0.042 (SE = 0.031), which is statistically indistinguishable from zero. At first, this appears consistent with DLR's null result seen in Table 2. However, the pre-test of parallel trends is decisively rejected ($p = 0$). The dynamic aggregation reveals significantly positive pre-treatment coefficients at long horizons, driven largely by the 2004 cohort, which had substantially higher employment growth than not-yet-treated counties throughout the early 1990s. This is reflective of the spatial heterogeneity DLR documents: high-minimum wage states grew faster than low-minimum wage states for reasons unrelated to minimum wage policy.

At the cohort level, the results reveal that the aggregate null hides heterogeneity. The 1992 cohort shows a persistent and significant negative employment ATT of -0.071 (SE = 0.022), contaminated by its large positive pre-trends. The 2004 and 2005 cohorts show effects near zero (-0.013 and -0.000), with flat pre-trends. The Goodman-Bacon decomposition further shows that the TWFE estimate using clean comparisons only is more negative than the full TWFE, consistent with the forbidden comparisons pulling the estimate in opposing directions (Figure 1).

These results validate DLR's methodological contribution. Even with a heterogeneity-robust estimator, covariate adjustment, and a clean not-yet-treated control group, the parallel trends assumption fails in the national sample. Spatial heterogeneity in employment trends is a geographic problem that cannot be resolved by modern difference-in-differences methods alone.

A natural extension (that was attempted) would be to partial out pair-period fixed effects via within-pair demeaning before applying the CS estimator (the Mundlak approach), which would combine DLR's local identification with CS's heterogeneity-robust weighting. When we attempted this method, the pre-test p-value rises to 1 after demeaning, which suggests parallel trends is no longer rejected. However, this result is not informative: the within-pair demeaning removes almost all identifying variation in the outcome. With only two counties per pair, demeaning within pair-year leaves only the within-pair differential. The CS estimator requires sufficient cross-sectional variation to fit the propensity score and outcome regression models, and the demeaning transformation collapses this variation before estimation begins. The CS results confirm that standard TWFE (without pair-period FEs) is contaminated by spatial heterogeneity in the national sample. DLR's Spec 6 is not vulnerable to this critique because the pair-period FEs effectively restrict comparisons to within-pair variation. This eliminates the spatial confound that drives the pre-trend failures in the national CS estimator. The contamination we document is a property of the all-county TWFE, not the preferred Specification 6.

Table 4: Wide Replication: CS (2021) Doubly-Robust DiD vs. DLR TWFE

	ATT	Std. Error
Panel A: DLR TWFE (Narrow Replication)		
Spec 1 — All counties, period FE	−0.177*	(0.097)
Spec 6 — CBCP, pair×period FE	+0.017	(0.054)
Panel B: CS (2021) DRDID, All-County Sample		
Overall ATT (simple)	−0.042	(0.031)
Dynamic ATT (post-treatment average)	−0.068*	(0.021)
Cohort 1992 ($n = 183$ counties)	−0.071*	(0.022)
Cohort 2004	−0.013*	(0.003)
Cohort 2005	−0.000	(0.008)
Goodman-Bacon clean comparisons only	−0.054	—
Goodman-Bacon TWFE (all comparisons)	−0.025	—
Pre-test p-value (parallel trends)	0.000	

Notes: Panel A reproduces DLR (2010) narrow replication results from Table 2. Panel B reports doubly-robust group-time ATT estimates from Callaway & Sant'Anna (2021) applied to the all-county annual panel (1,380 counties, 1990–2005). Treatment is a binary absorbing indicator equal to one from the first year a county's state minimum wage exceeded the federal floor. Three cohorts are identified: 1992 ($n = 183$), 2004, and 2005. Always-treated counties ($n = 379$, first treated at sample start) are excluded. Not-yet-treated counties serve as the control group. Covariates: log county population and log baseline total private employment. SEs clustered at state level. The Goodman-Bacon decomposition decomposes the TWFE estimate into clean (never/not-yet-treated) and timing-contaminated comparisons. Pre-test p-value tests the joint null of zero pre-treatment ATT(g,t) coefficients. Significance: * $p < 0.10$.
Note on comparability: CS and Goodman-Bacon use a binary absorbing treatment on an annual panel. DLR Specs 1 and 6 use continuous $\ln(MW)$ on a quarterly panel with county population and private sector controls. Magnitudes are not directly comparable but directional inference is.

Table 5: CS (2021) Dynamic ATT by Event Time — All-County Sample

Event time	Estimate	95% Band	Event time	Estimate	95% Band
Pre-treatment					
−15	+0.128	[−0.033, +0.289]	−7	+0.013	[−0.043, +0.068]
−14	+0.103	[−0.001, +0.207]	−6	+0.003	[−0.054, +0.061]
−13	+0.074	[−0.019, +0.167]	−5	−0.009	[−0.072, +0.054]
−12	+0.041	[−0.048, +0.130]	−4	−0.016	[−0.066, +0.035]
−11	+0.037	[−0.059, +0.133]	−3	−0.001	[−0.028, +0.027]
−10	+0.028	[−0.056, +0.111]	−2	+0.004	[−0.015, +0.022]
−9	+0.031	[−0.033, +0.095]	−1	—	—
−8	+0.024	[−0.034, +0.083]			
Post-treatment					
+0	−0.005	[−0.016, +0.007]	+8	−0.112*	[−0.169, −0.055]
+1	−0.026	[−0.058, +0.006]	+9	−0.106*	[−0.171, −0.042]
+2	−0.083*	[−0.121, −0.045]	+10	−0.090*	[−0.155, −0.025]
+3	−0.063*	[−0.103, −0.024]	+11	−0.049	[−0.110, +0.012]
+4	−0.089*	[−0.132, −0.046]	+12	−0.051	[−0.111, +0.008]
+5	−0.087*	[−0.144, −0.029]	+13	−0.049	[−0.123, +0.024]
+6	−0.060	[−0.125, +0.006]			
+7	−0.083*	[−0.151, −0.016]			

Notes: Dynamic ATT estimates from Callaway & Sant'Anna (2021), aggregated across cohorts by event time. Reference period is $t = -1$. Pre-treatment coefficients at $t - 14$ through $t - 8$ are positive and several are significant, indicating systematic pre-existing employment growth differentials between treated and not-yet-treated counties. This is consistent with DLR's diagnosis of spatial heterogeneity in employment trends. The pre-test of parallel trends is rejected ($p = 0$). Post-treatment coefficients are driven by the 1992 cohort; 2004 and 2005 cohorts show effects near zero. 95% simultaneous confidence bands. * band does not cover zero.

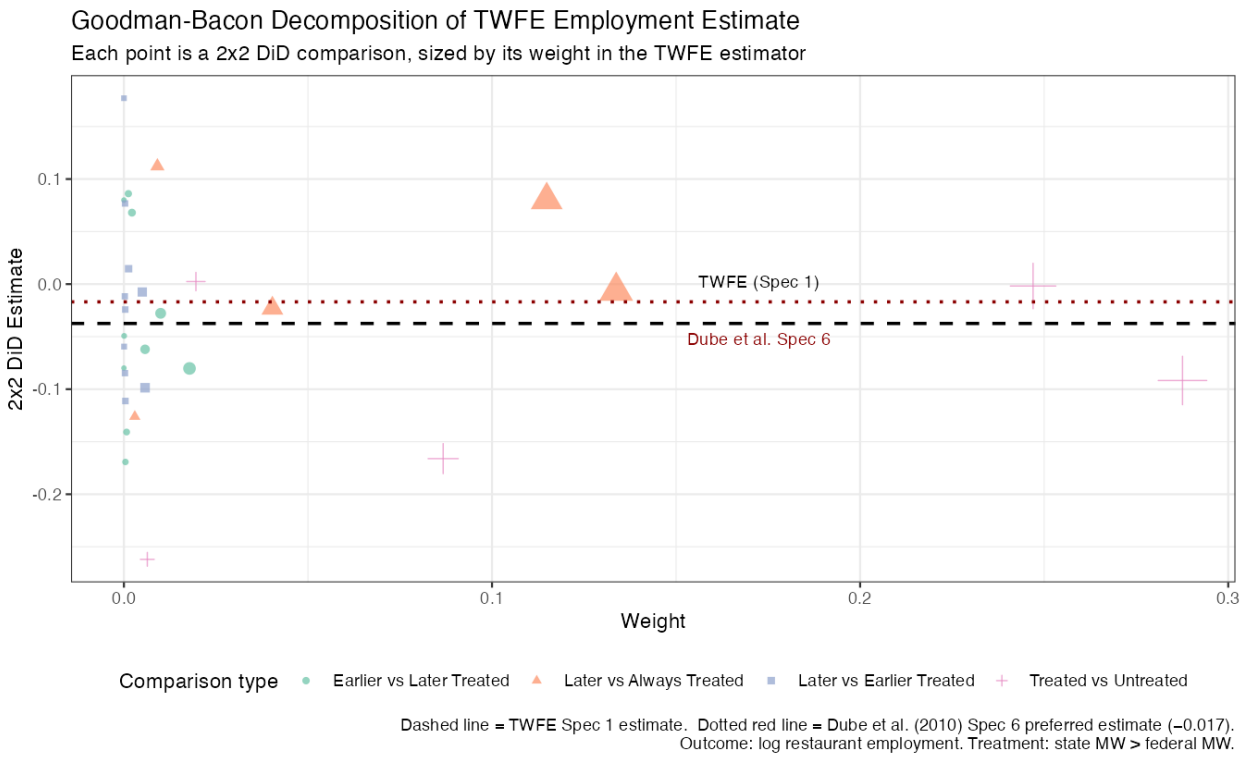

Figure 1

V Conclusion

This paper conducts a narrow and wide replication of Dube et al. (2010). The narrow replication confirms their core findings: earnings elasticities are stable at ≈ 0.20 across all specifications, and the employment elasticity falls to zero as a border-pair design is implemented.

The wide replication applies the doubly-robust estimator of Callaway and Sant'Anna (2021) to the full national sample. Parallel trends is decisively rejected even with heterogeneity-robust weighting and covariate adjustment, as large positive pre-treatment coefficients confirm that high-minimum-wage states had faster employment growth for reasons unrelated to minimum wage policy. These failures are a property of the national identification strategy. Pair-period fixed effects absorb the spatial confound that no amount of modern reweighting can substitute for. DLR's border-pair design performs well for the application.

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